

Source: Local Subject Media

24CS206 MACHINE LEARNING

Hours Per Week :	L	T	P	SL	C
	2	2	2	2	4

PREREQUISITE KNOWLEDGE: Probability & Linear Algebra, Python language.

COURSE DESCRIPTION AND OBJECTIVES:

This course introduces fundamental machine learning concepts, including supervised and unsupervised learning (clustering and dimensionality reduction). It helps students understand key challenges and select appropriate model complexity. Through practical case studies in areas such as computer vision, medical imaging, audio, and text, students gain real-world exposure. The lab component focuses on hands-on implementation using Python libraries like Scikit-learn and Matplotlib.

MODULE-1

UNIT-1 **6L+6T+6P+6SL=24 Hours**

INTRODUCTION

What is machine learning? Machine learning applications; Types of Learning: Supervised learning, unsupervised learning, and reinforcement learning. (T1,pp.1-9)

Model Training Essentials: Bias–Variance Trade-off (T2,pp. 67–69), Hypothesis Testing, Re-sampling methods: Sub sampling and Upsampling, SMOTE; Cross Validation (validation set, Leave-One-Cut (LOO), k-fold strategies),(R2,pp.198-215); Evaluation Metrics-Error functions, Confusion Matrix, Accuracy, Precision and Recall, F1 Score (R3,pp.86-89).

Regression Analysis: Linear Regression - Simple and Multiple Linear Regression (R2, pp.63-92), Polynomial Regression (R1, pp.290), Logistic Regression (R1,pp-119-127). Model Shrinkage-Ridge, and LASSO regression (R2, pp.237-241).

UNIT-2 **6L+6T+6P+6SL=24 Hours**

FEATURE SELECTION

Feature Selection Strategies: Types of feature selection Filter method: Chi-Square test, Wrapper method: forward/backward selection algorithms, Embedded method: Elastic net.(R3,pp.38-47)

Dimensionality Reduction: Matrix factorization(T1, pp.561-569), Principal components analysis, Linear discriminant analysis (T3, pp.207-226).

PRACTICES:

Exercise 1: ML Pipeline Construction

• Dataset: Any tabular dataset (e.g., student performance / housing)

Tasks:

• Identify problem type (classification/regression)
• Perform preprocessing (missing values, encoding, scaling)
• Split into train–test sets

Analysis Task: Justify chosen learning type (supervised/unsupervised)

Exercise 2: Cross-Validation Techniques

Tasks:

• Implement:
• Simple validation
• k-fold (k=5,10)
• Leave-One-Out (LOO)
• Compare accuracy

Analysis Task: Trade-off between computational cost and reliability

Exercise 3: Handling Imbalanced Dataset

• Dataset: Fraud detection / medical dataset

Tasks:

• Apply:

• Undersampling

• SMOTE

Analysis Task: Compare impact on recall and precision

Exercise 4: Linear vs Polynomial Regression

Tasks:

• Fit linear regression

• Fit polynomial regression (degree 2,3)

• Compare RMSE

• Train:

• Simple model (underfitting)

• Complex model (overfitting)

• Plot training vs validation error

Analysis Task: Identify overfitting, and bias vs variance issues

Exercise 5: Multiple Linear Regression

Tasks:

• Train model with multiple features

• Interpret coefficients

Analysis Task: Which features are most influential?

Exercise 6: Logistic Regression

Tasks:

• Binary classification using logistic regression

• Multi-class using multinomial regression

• Train model

• Compute:

• Accuracy, Precision, Recall, F1-score

• Plot confusion matrix

Analysis Task: Compare decision boundaries

Exercise 7: Regularization (Ridge vs LASSO)

Tasks:

• Train:

• Ridge regression

• LASSO regression

• Compare coefficients

Analysis Task:

• Which features are eliminated by LASSO?

Exercise 8: Feature selection Techniques

Tasks:

• Apply Chi-square test

• Rank features

• Implement:

• Forward selection

• Backward elimination

• Train Elastic Net model

• Analyze feature importance

Analysis Task: Select top-k features and justify

• Compare with Ridge & LASSO

SKILLS:

- ✓ *Statistical data analysis.*
- ✓ *Classify / Cluster data.*
- ✓ *Tool usage for developing ML applications.*

Exercise 9: LDA vs PCA

Tasks:

• Apply PCA and LDA on same dataset

• Visualize projections

Analysis Task:

• Which gives better class separation and why?

Exercise 10: End-to-End Model Optimization

• Dataset: Real-world dataset

Tasks:

• Perform:

• Feature selection

• Dimensionality reduction

• Model training (Regression/Classification)

• Evaluate using cross-validation

Analysis Task:

• Which combination gives best performance?

MODULE-2

UNIT-1 **6L+6T+6P+6SL=24 Hours**

CLASSIFICATION: Classification Types: Binary, Multi-class, and Multi-label Classification, (T2 pp.254-257) Decision tree induction - attribute selection measures; Tree pruning; CART (Classification and Regression Trees) (T2 pp.216), k-NN (k-Nearest Neighbor classifier) (T3 pp 64-66) and Fuzzy k-NN, (R3, pp.219).

UNIT-2 **6L+6T+6P+6SL=24 Hours**

ENSEMBLE METHODS: Voting: Soft Voting, Hard Voting, Bagging: Random Forest Trees, Boosting: Gradient Boosting, XGBoost, Stacking (T3, pp.183-202).

UNIT-3 **6L+6T+6P+6SL=24 Hours**

Clustering Techniques: Hierarchical methods; Agglomerative and divisive method, Fuzzy C-Means Clustering (R4, pp.578-581), Spectral Clustering (R5. Pp.396-413), Cluster Evaluation Methods (R4, pp.423-430).

PRACTICES:

Exercise 1: Binary Classification Pipeline

• Dataset: Breast Cancer / Spam dataset

Tasks:

• Perform preprocessing (missing values, normalization)

• Train a Logistic Regression / Decision Tree classifier

• Evaluate using Accuracy, Precision, Recall, F1-score

• Plot confusion matrix

Analysis Task: Interpret false positives vs false negatives

Exercise 2: Multi-class Classification

• Dataset: Iris / Digits

Tasks:

• Implement multi-class classification using k-NN and Fuzzy k-NN (assign membership values)

• Compare classification result with k = 3, 5, 7

• Plot decision boundaries

Analysis Task: Explain effect of k on bias-variance, When does fuzzy approach outperform k-NN?

Exercise 3: CART Implementation

Tasks:

• Train Decision Tree using Gini Index

• Visualize tree structure

• Apply pruning

Analysis Task: Compare overfitting before/after pruning

Exercise 4: Voting Classifier

Tasks:

• Train 3 models (Logistic Regression, k-NN, Decision Tree)

• Combine using:

• Hard Voting

• Soft Voting

Analysis Task: Which voting performs better and why?

Exercise 5: Random Forest

Tasks:

• Train a Decision Tree and Random Forest

• Compare accuracy and variance

Analysis Task: Explain why Random Forest reduces overfitting

Exercise 6: Boosting Techniques

Tasks:

• Train Gradient Boosting / XGBoost

Analysis Task:

• Compare performance across models

• Identify cases where boosting fails

Exercise 7: Stacking

Tasks:

• Use 2 base learners + 1 meta-learner

• Compare with individual models

Analysis Task: Explain role of meta-learner

Exercise 8: Fuzzy C-Means Clustering

Tasks:

• Implement FCM algorithm

• Visualize membership values

• Compute:

• Silhouette Score, Davies-Bouldin Index

Analysis Task: Compare with K-Means clustering

Exercise 9: End-to-End ML Comparison

• Dataset: Any real-world dataset (e.g., healthcare/customer data)

Tasks:

• Apply:

• Classification (k-NN, CART)

• Ensemble (Random Forest, XGBoost)

• Clustering (FCM / Spectral)

• Compare results

Analysis Task:

• When to use classification vs clustering?

COURSE OUTCOMES:

CO No.	Course Outcome	Bloom's Level	Module No.	Mapping with POs
C1	Apply a wide variety of learning algorithms such as Probabilistic, Discriminative and Generative algorithms for a given application.	Apply	Module 1 Unit 1, Module 2 Unit 1 & 2	PO1, PO2, PO3
C2	Design an end-to-end Machine-learning model to realize solutions for real-world problems.	Create	All Modules	PO1, PO2, PO3, PO5

CO No.	Course Outcome	Bloom's Level	Module No.	Mapping with POs
C3	Implement various machine learning models using advanced ML tools.	Apply	Module 1 Unit 1 & 2, Module 2 Unit 1, 2, 3	PO2, PO3, PO4
C4	Analyze and evaluate the performance of various machine learning model approaches on different kinds of data.	Analyze& Evaluate	Module 1 Unit 1, Module 2 Unit 3	PO1, PO2, PO4
C5	Develop expertise in ensemble methods and their application in improving predictive performance.	Apply	Module 2 Unit 2	PO2, PO3, PO4

MAPPING OF SDGS AND IKS

Unit	Topic	Sustainable Development Goals (SDGs)	Indian Knowledge System (IKS)
Unit 1	Introduction to Machine Learning and Model Training Essentials	SDG 4 – Quality Education SDG 9 – Industry, Innovation & Infrastructure	Emphasizes logic, pattern recognition, and reasoning—core to traditional Indian pedagogy.
Unit 2	Feature Selection and Dimensionality Reduction	SDG 9 – Industry, Innovation & Infrastructure	Reflects ancient systems of classification and decision-making in Ayurveda and Nyaya system.
Unit 3	Classification Techniques (CART, k-NN, Fuzzy k-NN)	SDG 3 – Good Health & Well-being SDG 4 – Quality Education	Use of heuristic knowledge and fuzzy logic resonates with decision-making in traditional texts.
Unit 4	Ensemble Methods	SDG 8 – Decent Work & Economic Growth SDG 9 – Innovation	Aligns with collective intelligence concepts in ancient Indian systems like Panchayat decision models.
Unit 5	Clustering Techniques (Fuzzy C-Means, Spectral Clustering)	SDG 11 – Sustainable Cities SDG 13 – Climate Action	Community grouping, pattern formation reflect social structures in Indian knowledge history.

TEXT BOOKS:

REFERENCE BOOKS: